ChurnGuard | Customer Churn Prediction

Milestone 2 — Model Development, Optimization & Business Impact

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Executive decision brief

ChurnGuard commissioned an AI-powered, cross-industry system to identify accounts at risk of churn and enable targeted retention. Among the models evaluated, the tuned Random Forest achieved the strongest cross-validation Average Precision (97.52%). On 2,252 held-out accounts, it identified 355 of 379 churned accounts with 95.69% precision and 93.67% recall. These results support an exploratory pilot, not immediate production deployment, because the timing of the churn outcome relative to source features is undocumented.

TEST ACCURACY	TEST PRECISION	TEST RECALL	TEST AVERAGE PRECISION
98.22%	95.69%	93.67%	98.70%

1 | Business setting and data basis

ChurnGuard aims to help client retention, marketing and customer-service teams prioritize outreach; finance teams assess campaign economics; and data science/product teams manage model quality. The dataset contains 11,260 unique accounts, of which 1,896 (16.84%) are labeled churned. Each row represents one account, which may contain multiple users.

The data dictionary describes a 12-month lookback for complaints, customer-care contacts, coupon usage, average monthly revenue and cashback; revenue growth compares the most recent 12 months with months 13–24. Exact calendar dates, data-capture cadence, churn definition and outcome horizon remain unspecified.

2 | Model development and evaluation design

An 80/20 stratified split allocated 9,008 accounts to model development and 2,252 to final testing. Training-fold preprocessing handled missing values, categorical encoding and numerical scaling; AccountID was excluded. Five-fold stratified cross-validation compared Logistic Regression, Decision Tree and Random Forest. RandomizedSearchCV tuned Decision Tree and Random Forest (30 candidates each; 150 fits per search). Average Precision was the primary selection metric because churn is the minority class; recall, precision, F1 and ROC-AUC provided supporting evidence.

3 | Model comparison — cross-validation

Model	Accuracy	Precision	Recall	F1	Avg. Precision
Logistic Regression	87.91%	76.39%	40.87%	53.23%	65.61%
Decision Tree	93.77%	82.09%	80.62%	81.35%	69.47%
Tuned Decision Tree	89.20%	62.76%	88.27%	73.35%	79.74%
Random Forest	96.37%	97.18%	80.82%	88.24%	97.24%
Tuned Random Forest	97.25%	92.48%	91.10%	91.78%	97.52%

Cross-validation averages are development estimates; tuned-model estimates reuse folds used for parameter selection and may be optimistic.

What tuning changed

Tuning Random Forest improved recall from 80.82% to 91.10% (+10.28 percentage points) and F1 from 88.24% to 91.78%, while precision declined from 97.18% to 92.48%. Average Precision rose modestly from 97.24% to 97.52%. Decision Tree tuning improved Average Precision from 69.47% to 79.74%, but reduced precision and F1. The tuned Random Forest therefore provided the strongest overall development-stage balance.

The chosen Random Forest used 200 trees, balanced class weights, log2 feature sampling and unrestricted maximum depth. Its training scores remained nearly perfect; validation recall was 91.10% versus 100% on training folds. This indicates a remaining generalization gap, even though the validation results are strong.

4 | Final held-out test results

Measure	Result	Business interpretation
Accuracy	98.22%	2,212 of 2,252 accounts correctly classified
Precision	95.69%	355 of 371 flagged accounts were churned
Recall	93.67%	355 of 379 churned accounts detected
F1-score	94.67%	Strong precision-recall balance
ROC-AUC	99.69%	Strong ranking discrimination
Average Precision	98.70%	Strong precision-recall ranking on this test sample

Confusion matrix — actual outcomes versus predictions

Actual / Predicted	Predicted retained	Predicted churned
Actually retained	1,857 (true negative)	16 (false positive)
Actually churned	24 (false negative)	355 (true positive)

For an operational team, the 24 missed churners represent possible missed outreach opportunities, while the 16 false alerts represent potentially unnecessary contacts. These counts reflect the test sample, not an estimate of prevented churn or financial return.

5 | What drives predictive performance?

Permutation importance was calculated on a validation subset of the training data. The scores represent the mean reduction in Average Precision after shuffling each feature; they are not causal effects or percentages of churn explained.

Feature	Importance	Connection to observed churn patterns
Tenure	0.2799	Accounts with tenure 0–1: ~51.9% churn
Complaint status	0.1585	Complaints: 31.76% churn vs 11.14% without
Marital status	0.0821	Predictive signal; fairness review required
Account segment	0.0809	Regular Plus: 27.33% churn
Days since care contact	0.0542	Customer-care recency contributes signal

The clearest combined EDA finding is that early-tenure accounts (0–1) with a complaint had 74.62% observed churn. Regular Plus accounts with complaints had 53.84% churn. These are priorities for investigation and controlled intervention testing, not proof that outreach will cause retention.

6 | Business recommendations

Priority	Recommended action	Success measure
Early account experience	Offer onboarding check-ins and targeted help for new accounts	Retention uplift in a controlled pilot
Complaint recovery	Resolve complaints quickly; follow up after closure, especially for new accounts	Resolution time; incremental retention
Segment-led outreach	Investigate Regular Plus needs and tailor outreach rather than blanket discounts	Incremental contribution margin
Risk-based workflow	Use validated risk scores to prioritize service teams; set thresholds using validation economics	Recall, precision, cost per contact

7 | Practical rollout and governance

Stage	Gate before progression
1. Data validation	Define feature cutoff, churn event and prediction horizon; validate tenure=99 and extreme values
2. Temporal validation	Test on later-dated accounts; assess drift, leakage and subgroup performance
3. Controlled pilot	Randomize eligible at-risk accounts to intervention/control; estimate uplift and costs
4. Operational rollout	Integrate scores into CRM with human oversight, monitoring and retraining policy

8 | Limitations and executive conclusion

Temporal validity: historical feature windows are known, but the reference date and churn outcome period are not; post-outcome information may be present. Generalization: random-split test results do not prove future or cross-industry performance. Data quality: extreme tenure, revenue and cashback values require business verification. Fairness: demographic predictors such as marital status and gender require governance review. Business value: no intervention uplift, campaign costs or realized savings have been measured.

Recommendation: retain tuned Random Forest as the leading exploratory candidate, complete temporal and leakage checks, then run a measured retention pilot. ChurnGuard has demonstrated promising predictive discrimination; the next decision is whether the system can predict future churn reliably and generate positive incremental business value.